



OFFICIAL OWASP COMMUNITY PROJECT

OWASP Quantum Security Project

Project kick-off call

Prepare now. Protect tomorrow.

Tuesday 30 June 2026

Co-leads: John Sotiropoulos and Roy Barkay

THE PROBLEM SPACE

Security planning must already account for quantum-capable adversaries

NIST, UK NCSC, and EU institutions have issued concrete guidance - post-quantum cryptography, crypto-agility, and harvest-now-decrypt-later mitigation - while quantum systems move from research into cloud and enterprise environments.

01

No concise, prioritised view of quantum security risks

02

No widely adopted threat model for quantum platforms

03

Organisational readiness and platform security treated separately

WHY THIS PROJECT

The antidote to hype and fear

A practical, vendor-neutral community effort that maps quantum risk in a systematic, risk-based, actionable way.



Practical and immediate

Help defenders act on quantum risk today, not in a decade.



Systematic and risk-based

Explore and map the future in a structured, prioritised way.



Actionable and grounded

Tied to recognised guidance from NIST, NCSC and the EU.



One community

Connecting research, industry and practitioners in a shared effort.

HOW WE WILL WORK

Start lean, scale with the community



Governance

- Start lean; scale as the work takes shape
- Working group leads and a small core group as we define scope
- **Project Charter to be published next week**



Cadence

- Monthly community call to begin
- Additional calls on demand as the work picks up



Leads

John Sotiropoulos

Roy Barker

Expert-backed, Community Driven:

Open transparent. Peer reviews –
internal, community public
Project Lead Signoff

WORK PLAN

Two Tracks

Track 1

Primary · Year 1

Quantum security risks & readiness

Delivers an OWASP Top 10 for Quantum Security Risks, mitigation guidance, and a Quantum Readiness Assessment Assistant.

2 deliverables

Track 2

Parallel · Secondary

Quantum platform threat modeling

Defines threat models, attack surfaces, trust boundaries, and secure design guidance for quantum platforms.

4 deliverables

WHERE WE BEGIN

OWASP Top 10 for Quantum Security Risks

Our starting point, not the finished list. A bootstrap to open the conversation.

v0.1 Today - bootstrap



v0.2 Entry leads onboard



v1 Oct/Nov 2026



How it works

- Suggest and discuss candidate risks via the submission form
- Work moves to GitHub as the project matures
- Seeking entry leads as we move to v0.2

Submission form for initial submissions

Success criteria

- ✓ Useful to defenders today
- ✓ Actionable and practical
- ✓ Linked and grounded to existing guidance (NIST, NCSC, EU)

OWASP Top 10 for Quantum Security Risks

At a glance — Draft v0.1 — June 2026

Migration surface (QS01-QS07)

Platform surface (QS08-QS10)

QS01

Harvest-Now-Decrypt-Later

Adversaries are capturing encrypted traffic and stored ciphertext today, expecting a CRQC will allow retroactive decryption. Concrete now, not contingent on quantum hardware availability.

Anchor: EU Roadmap end-2030; NCSC 2031; NSM-10

QS02

Long-Lived Sensitive Data

Data with confidentiality requirements past CRQC arrival cannot be protected by today's public-key primitives alone. Mosca's inequality ($X+Y>Z$) is the operative planning frame.

Anchor: Mosca's inequality; EU end-2030

QS03

Vulnerable Signatures and Code-Signing

RSA, ECDSA, EdDSA signatures break under Shor. Code-signing, supply-chain integrity, certificates, and long-term non-repudiation are exposed once a CRQC exists.

Anchor: CNSA 2.0 by 2030; CRA Annex I

QS04

Absent Cryptographic Inventory

Organisations cannot migrate cryptography they have not catalogued. Absence of a structured CBOM is the single most common blocker to PQC migration in 2026.

Anchor: NCSC 2028; EU Roadmap end-2026

QS05

Crypto-Agility Failures

Hard-coded algorithms, frozen vendor stacks, and brittle protocols prevent response to standards updates and parameter changes. Crypto-agility is itself a security control.

Anchor: NIS2 Article 21(2)(h); IETF PQUIP

QS06

Insecure Migration and Hybrid Misuse

Silent fallback to classical, broken hybrid KEMs, downgrade attacks, weak parameters, and non-constant-time implementations. Documented by IETF PQUIP and TLS WG.

Anchor: EU Roadmap end-2030 standalone-classical prohibition

QS07

Hardware Roots of Trust

TPMs, UEFI keys, smart cards, embedded device certificates, and OT controllers have the longest replacement cycle. Devices in service today will outlive any practical PQC deadline.

Anchor: NCSC 2028; CRA Annex IV

QS08

QPU Tenant Isolation Failures

Crosstalk between qubits degrades victim computations on shared QPUs (Choudhury NDSS 2025). Reset gates leak state across tenant boundaries (Xu CCS 2023). Shared-QPU isolation requires evidence.

Anchor: Choudhury NDSS 2025; Xu CCS 2023

QS09

Toolchain and Compiler Compromise

Circuit theft via compromised compilers (Suresh HASP 2021). QTrojan disables data encoding by manipulating pulse calibrations disguised as routine (Chu 2023). Quantum supply-chain risk.

Anchor: Suresh HASP 2021; Chu 2023

QS10

Side-Channel and Control-Plane Exposure

Timing analysis of resets reveals execution patterns (Mi CCS 2022). Power traces reverse-engineer gate-level circuits (Xu CCS 2023). Control-plane compromise is workload compromise.

Anchor: Mi CCS 2022; Xu CCS 2023

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Quantum Readiness Assessment Assistant

An operational
assessment
description

Operational Support

**Not Just
Content**

- Guides structured self-assessment
- Maps exposure to the Top 10 risks
- Generates prioritised mitigations
- Provides transparent scoring logic
- Vendor-neutral & community-reviewed

Quantum platform threat modeling

Documenting how quantum computing platforms and hybrid quantum-classical architectures can be attacked - and how to design them securely.

WHAT IT DEFINES

- ♦ Threat models for quantum platforms
- ♦ Trust boundaries & control planes
- ♦ Secure design guidance
- ♦ Hybrid architecture attack surfaces

DELIVERABLES

1. Quantum Platform Threat Model
2. Reference architecture diagrams
3. Attack surface mapping
4. Secure design & mitigation guidance

Threat Modelling Will Unveil New Areas



Let's review early areas that matter

- Quantum-backed Adversarial AI
- Multi-tenant QPU isolation and crosstalk side channels
- Quantum cloud access and QPU-as-a-ServiceTrust boundaries, control planes and platform design
- Calibration and noise-data integrity
- Quantum software stack: SDKs, compilers and transpiler
- Quantum networking and QKD
- Supply chain and Quantum hardware, firmware



Propose a topic

Have a topic of interest? Submit it via the form and help shape what we build next.

Topic form: [add link](#) before the call

The roadmap at a glance

Q2 2026

LAUNCH

- Kick-off and onboarding
- Scope, terminology, structure
- Agree Working Groups
- Map authoritative guidance

Q3 2026

BUILD

- Draft Top 10 candidates
- Community review
- Build assessment assistant
- Formalize Track 2 Working Groups

Q4 2026

PUBLISH

- Publish v1 Top 10 (Oct/Nov)
- Release assessment assistant
- Initial platform threat model

2027

EVOLVE

- Community refinement
- Align mitigation patterns
- Deepen Track 2; next phase

Have a voice

<https://forms.gle/8NbEEX6mmiKdUXxXA>



Top 10 for Quantum Security

<https://forms.gle/n7BicJJpQJFQ8eRz7>



Work Areas and Proposals

GET INVOLVED

Join us in securing the quantum frontier



Homepage

quantum-owasp.org



GitHub

github.com/OWASP/quantum-security-project



OWASP Slack

[#project-quantum-security](#)



Email

contribute@quantum-owasp.org

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